Ribhu Hooja

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Education

Dartmouth College, Hanover, NH **June 2027**
Bachelor of Arts: Major in Mathematics, Minor in Computer Science **GPA 3.93/4.0**
Citations for meritorious performance (added to 2% of awarded grades): 10

Papers

τ and ϵ invariants of generalized Mazur patterns (with S. Kim, F. Rupchin, S. Sinha, and A. Tsypin).
In preparation; preprint available upon request.

Research Experience

SHUR REU, Dartmouth Summer 2026
Mentors: Ina Petkova and Abhishek Mallick

- Computed τ and ε for a family of pattern knots using bordered Floer homology
- Wrote software for the bordered Floer computations, available on GitHub.

Park City Mathematics Institute June 28 – July 18, 2026
Knotted Surfaces in Four-Manifolds

- Attended the Undergraduate Summer School and participated in an Experimental Math Lab project on knot Floer homology of (1,1)-knots.

Spectral geometry research, Dartmouth June 2025 – present
Senior thesis; Presidential Scholar (2025–26); UGAR leave-term grant (Summer 2025). Mentor: Craig Sutton

- Studying the problem of determining a closed locally homogeneous 3-manifold's Thurston geometry from its Laplace spectrum using heat invariants
- Working on showing that certain Berger lens spaces are not isospectral using the Duistermaat–Guillemin trace formula

Talks

Satellite operations and bordered Heegaard Floer homology, Dartmouth Topology Seminar September 24, 2026
Sunada's method: using group theory to construct isospectral drums, Hudson River Undergraduate Math Conference April 25, 2026

Teaching

(upcoming) Teaching assistant, Math 104 Differential Topology (Ina Petkova) Winter 2027
Grader, Math 72.01 Calculus on Manifolds (Craig Sutton) Spring 2026
Teaching assistant, COSC 31 Algorithms (Deeparnab Chakrabarty) Summer 2025
Section leader, COSC 10 Problem Solving via Object-Oriented Programming (Timothy Pierson) Fall 2024

Honors and Awards

Jack Byrne Scholarship in Math and Society ($10,000 per year) 2025–2027
Thayer Math Prize, 2nd place 2024

Other work

Trust Lab Spring 2025
URAD Scholar. Mentor: Sergey Bratus
Program verification in Lean; attended Summer School on Formal Techniques, SRI, May 24–30, 2025

DALI Lab August 2024–March 2025
AR/VR, Data science
ML-assisted cancer-detection microscope (Python, AWS); 3D control-flow visualizer (Unity/C#)

LISP Lab January–August 2024
UGAR researcher (Summer 2024); FYREE researcher (January–May 2024). Mentor: Peter Chin
Embedding graphs into hyperbolic space with machine learning for faster routing

Skills

Programming: Python, C, C#, Java; LaTeX
Languages: English, Hindi